

Vellore Institute of Technology
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INSURECHECK- NLP : Automated Insurance Policy Claim Exclusion and Hidden Clause Extraction Engine Using Fine-Tuned Transformer Architecture

PROJECT REVIEW 1 REPORT

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Project Guide

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ABSTRACT

Insurance policy language is dry, dense and often uses legal jargon, exclusionary clauses and other insufferable language to narrow the scope for insurance coverage. A common occurrence is policyholders being denied coverage for claims when there is fine print that they are unaware of, including room-rent sub-limits, multi-year waiting periods, copayment requirements, and lists of non-pays. This project introduces an intelligent framework for Natural Language Processing (NLP) to automatically identify, categorize and quantify claim exclusions and risk traps in insurance policy PDFs, named InsureCheck-NLP. The proposed pipeline integrates pdfplumber for section-aware text extraction, optimized DeBERTa-v3 to perform sequence labeling and clause classification, and SentenceTransformers (all-MiniLM-L6-v2) to map the text to a standardized insurance risk taxonomy. InsureCheck-NLP assigns a composite Policy Transparency & Claim Risk Score (%) and provides an easy-to-understand breakdown of uninsurable conditions. The system performs well on four major exclusion categories on public legal and insurance contract benchmarks (CUAD and custom insurance corpora), respectively, with strong F1-scores. An interactive Streamlit dashboard offers policyholders visual risk heatmaps and claim transparency analytics.

PROBLEM STATEMENT AND OBJECTIVE

Problem Statement

Insurance policies include extensive legal jargon in the form of long documents that are buried with many pages of PDF and a lot of critically important terms that disallow coverage (like disease waiting periods, co-pay percentages, cap limits, etc.). In complex legal language, traditional search tools work only with simple keywords (such as "insurer shall not be liable for pre-existing ailments prior to 36 months" vs. "3-year waiting period for past illnesses"). The need is an automated and domain specific NLP system that can extract hidden exclusion clauses and convert dense legal jargons into plain, actionable language.

Objectives (Max 2 for Review I)

Objective 1: To create and apply a contextual NLP pipeline using fine-tuned DeBERTa-v3 and SentenceTransformers to label and classify insurance claims exclusions, based on the four main categories: Waiting Periods, Sub-Limits, Non-Payable Items, and Co-Pay Rules.
Objective 2: Create an automated Claim Risk Scoring Algorithm, and build an interactive Streamlit UI for parsing policy PDFs, drawing attention to hidden risk traps, and providing plain-English summary for decision support

LITERATURE SURVEY TABLE :-

S. No.	Title of Paper, Publisher & Year	Problem Addressed	Algorithms Used	Metrics Used	Performance Attained	Limitations (Gap Identified)	Scope for Future Work
1	Technical evaluation of language models adapted for the automation of legal contracts: clause extraction, classification, and summarizationFrontiers in Artificial Intelligence, 2026	Automating legal contract processing (clause extraction, classification, summarization) in data-constrained domains.	Domain-adapted DeBERTa/BERT, HuggingFace Transformers, scikit-learn, seqeval	Macro-F1, Span-level F1, ROUGE-L, BERTScore	Macro-F1: 0.921 in classification, Span F1: 0.903 in clause extraction, ROUGE-L: 0.886	Tested on private-sector corporate contracts; lacks specific insurance policy claim risk scoring.	Adapting legal transformers to consumer insurance policies and multi-category risk weighting.
2	Insurance Policy Summarization Using Natural Language Processing (NLP)Frontiers in Health Informatics, 2024	High complexity and length of insurance policy terms hindering policyholder comprehension of exclusions.	Hybrid Extraction-based & Abstractive Summarization, Tokenization, POS Tagging, TextRank	ROUGE-1, ROUGE-L, Edit Distance, Cosine Similarity, Human Evaluation	High summary fidelity across coverage and exclusion terms; improved user understandability.	Basic extractive-abstractive pipeline lacks explicit fine-grained clause extraction and numeric risk scoring.	Incorporating fine-grained legal entity extraction and automated risk score calculation.
3	AI-powered decision-making in facilitating insurance claim dispute resolutionAnnals of Operations Research (Springer), 2023	Prolonged dispute resolution and subjective cost estimation in road traffic accident (RTA) insurance	Predictive Regression & Machine Learning Decision-Making Models, Asymptotic Injury Modeling	Mean Absolute Error (MAE), RMSE, Judicial Cost Estimation Accuracy	High predictive accuracy in judicial cost estimation across 88 real-life RTA case studies.	Focused on numerical/judicial dispute costs rather than unstructured text policy clause extraction.	Integrating text-mining of insurance policy exclusion clauses with financial dispute estimation.

		claims.					
4	Transforming Health Insurance Claims Adjudication using Blockchain TechnologyProcedia Computer Science (Elsevier), 2023	Inefficiencies, tampering risks, and delays in centralized health insurance claims adjudication.	Smart Contracts, Decentralized Ledger Technology (DLT), Automated Rules Verification	Claim Settlement Latency, System Security, Throughput (TPS)	Substantial reduction in claims settlement time from weeks to near real-time automated execution.	Focuses on blockchain architecture; lacks natural language processing for parsing raw policy text.	Combining NLP policy clause extraction with smart contract execution for automated claims.
5	Explainable AI and Law: An Evidential SurveyDigital Society (Springer), 2024	Black-box opacities in AI legal decision-making reducing accountability and transparency.	Explainable AI (XAI) Taxonomy Mapping, Evidential Reasoning Frameworks	Explainability Fidelity, Stakeholder Trust Score, Interpretability	Developed a structured taxonomy aligning legal inference types with transparent algorithmic models.	Conceptual survey framework without automated empirical text processing code.	Implementing XAI token-attribution heads (e.g. SHAP/Integrated Gradients) in legal NLP classifiers.
6	Legal Information Retrieval and Entailment Using Transformer-based ApproachesThe Review of Socionetwork Strategies (Springer), 2024	Information overload and complex legal entailment verification in case law and statute retrieval (COLIEE).	Sentence-Transformers, Legal-BERT-ECHR, Stacking-Voting Ensembles, Binary Classification	Precision, Recall, F1-Score, COLIEE Task Accuracy	Superior entailment accuracy using Legal-BERT-ECHR on legal paraphrasing and condition matching.	Evaluated on statute law databases; does not target consumer insurance policy contracts.	Adapting legal entailment transformers to premise-claim verification in insurance contracts.

7	InsuranceRAG: A multi-module retrieval-augmented AI system for intelligent insurance assistanceSoftware Impacts (Elsevier), 2026	Lack of transparent, conversational AI tools for query resolution and clause-level document retrieval in health insurance.	Retrieval-Augmented Generation (RAG), FAISS Vector Store, Domain-adapted Embeddings, LLM Evaluator	Retrieval Recall@K, Answer Faithfulness, Latency, User Evaluation	High retrieval accuracy and explainable clause-level responses via a web dashboard.	Focuses on conversational QA without calculating structured, composite policy claim risk scores.	Integrating automated policy risk scoring algorithms with RAG clause extraction.
8	Optimizing Claim Assessment Process for Property Insurance using Machine LearningProcedia Computer Science (Elsevier), 2024	Data imbalance and limited feature importance in property insurance claim frequency and risk prediction.	Random Forest, Logistic Regression, SMOTE Data Imbalance Oversampling	Recall, Precision, F1-Score	Achieved 96% Recall with Random Forest after applying SMOTE balancing.	Operates on structured tabular risk features rather than unstructured PDF contract text.	Combining structured tabular ML models with text-based NLP exclusion extraction.
9	Structure-Aware Uncertainty-Calibrated Evaluation for Legal Contract Knowledge Graph ExtractionProcedia Computer Science (Elsevier), 2026	Evaluating epistemic uncertainty and inter-judge disagreement in legal contract knowledge graph extraction.	Multi-Judge Ensemble (Semantic + Structural + LLM Judges), Isotonic Regression Calibration	Coefficient of Variation (CV), AUC for Error Detection, Calibration Error	Achieved 0.982 AUC for error detection in contract knowledge graph extraction.	High computational overhead of multi-judge ensembles; requires reference-free validation data.	Lightweight single-pass uncertainty calibration for real-time contract scanning.

10	A case study for automated attribute extraction from legal documents using large language modelsArtificial Intelligence and Law (Springer), 2026	Automated extraction of fine-grained concepts, attributes, and tags from unstructured legal proceedings.	Weakly Supervised Sequence Labeling, Few-Shot Prompting with LLaMA-2 & Black-Box LLMs	Span Extraction Precision, Recall, Macro F1, Downstream Prediction Accuracy	LLaMA-2 achieved superior few-shot attribute generation and downstream statute prediction accuracy.	LLM hallucination risk and prompt sensitivity in unconstrained few-shot prompting regimes.	Combining LLM few-shot extraction with fine-tuned transformer classification heads.
11	Health insurance fraud detection based on multi-channel heterogeneous graph structure learningHeliyon (Elsevier), 2024	Detecting complex, evolving health insurance fraud tactics across heterogeneous entities (patients, doctors, medicines).	Multi-channel Heterogeneous Graph Structured Learning (MHGSL), Graph Convolutional Networks (GCN), KNN	Classification Accuracy, Macro F1, Precision, Recall, AUC-ROC	High fraud detection accuracy by capturing multi-entity relational topology.	Requires structured relational graph data; does not process unstructured policy document text.	Graph-augmented text processing linking policy exclusions with claim history.
12	Addressing the notion of trust around ChatGPT in the high-stakes use case of insuranceTechnology in Society (Elsevier), 2024	Trust, governance, and financial inclusion risks around consumer-facing LLMs in high-stakes insurance applications.	AI Governance Evaluation Framework, Dread Risk & Trust Judgment Analysis	Qualitative Trust Metrics, Risk Governance Alignment	Established domain-specific governance framework for consumer-facing LLMs under the EU AI Act.	Conceptual commentary paper without empirical model implementation.	Developing verifiable, explainable NLP systems that adhere to EU AI Act transparency rules.

13	Legal Eligibility Inference from Text: Constraint Extraction with Pretrained Language ModelsProcedia Computer Science (Elsevier), 2025	Automated extraction of structured eligibility restrictions and condition-action pairings from legal text.	Pre-trained BERT, Named Entity Recognition (NER), Condition-Action Pairing Rules, Entitlement Logic	NER Precision, Recall, Constraint Extraction Accuracy	High precision in formalizing entitlement logic into machine-readable constraints.	Restricted to eligibility rules; does not calculate composite policy financial risk scores.	Extending constraint extraction to insurance policy claim exclusion weighting.
14	Software engineering meets legal texts: LLMs for auto detection of contract smellsMachine Learning with Applications (Elsevier), 2024	Detecting 'contract smells' (text patterns indicative of significant issues or ambiguities in legal contracts).	GPT-4 Semi-Automated Labeling, BERT, LEGAL-BERT Fine-Tuning, Single vs Multi-Class Classification	Macro F1, Accuracy, Single-Class vs Multi-Class Precision	High accuracy achieved with fine-tuned LEGAL-BERT for single-class contract smell detection.	Identifies general contractual ambiguity rather than specific consumer insurance exclusion traps.	Applying contract smell concepts to insurance policy fine-print exclusion traps.
15	Automated construction contract analysis for risk and responsibility assessment using natural language processing and machine learningAutomation in Construction (Elsevier), 2023	Automating contract review to extract risk and responsibility information for risk management plans.	NLP Vectorization, Machine Learning Text Classification, Ensemble Models, Sentence Type Classifier	Accuracy, Macro F1-Score	High sentence classification accuracy across Obligation, Risk, and Right categories (3,485 sentences).	Domain-specific to construction contracts (FIDIC); not evaluated on health/life insurance policies.	Cross-domain adaptation of risk/responsibility sentence classification to consumer insurance.

Paper 1: Technical evaluation of language models for automated legal contracts (clause extraction, classification and summarization) :-

Author(s): Toraiko Shimazaki, Mutsuki Hanzawa, and Tadamasa Uchitani

Paper 3012691, Frontiers in Artificial Intelligence, 2026.

Title: Water level declines in the Upper Floral City Pool, St. Lucie County, Florida, since 2009Abstract: A reduction in water level in the Upper Floral City Pool, St. Lucie County, Florida, since 2009 is documented.

(ii) Problem & Objectives: Legal Contract Processing Automation in Data-constrained Domains (Clause Extraction, Classification, Summarization)

(iv) Data Collected: Training data, test data, dev data.(v) Data Processed: none.

The metrics used for performance and limitations are: Macro-F1, Span-level F1, ROUGE-L, BERTScore.

Attained: Macro-F1: 0.921 (Macro-F1: classification), Span F1: 0.903 (Span F1: clause extraction), ROUGE-L: 0.886 (ROUGE-L: clause extraction):

Limitations: Applied to contracts in private sector, no specific insurance policy claim risk scoring.

Consumer insurance policies and multi-category risk weighting as future scope of adapting legal transformers.

Paper 2: Summarizing Insurance Policies with Natural Language Processing (NLP)

Summarizing Insurance Policies using Natural Language Processing (NLP) (i)

Health Informatics frontiers, 2024

Authors: Sapana Kolambe and Dr. Parminder Kaur

(ii) Problem & Objectives: Long and complex insurance policy terms are making comprehension hard for policyholders as they read through exclusions.

(iii) Methodology Adopted: Hybrid Extraction-based & Abstractive Summarization, Tokenization, POS Tagging, TextRank

(iv) Performance & Limitations: Metrics: ROUGE-1, ROUGE-L, Edit Distance, Cosine Similarity, Human Evaluation

Achieved: Good fidelity in coverage and exclusion terms and user understandability.

Constraints: Basic extractive-abstractive pipeline does not explicitly extract clauses and numeric risk scores.

Future Scope: Fine-grained legal entity extraction and automated risk score calculation.

PAPER 3 : Powered by AI, how to manage insurance claim disputes.

(i) Title, Author, Publisher: AI-powered decision-making in facilitating insurance claim dispute resolution

The paper is published in Annals of Operations Research (Springer).

Authors: Wen Zhang, Jingwen Shi, Xiaojun Wang, Henry Wynn

(ii) Problem & Objectives: The long time taken to settle road traffic accident (RTA) insurance claims and the difficulty of getting objective cost estimates.

(iii) Methodology Adopted: Predictive Regression & Machine Learning Decision-Making Models, Asymptotic Injury Modeling, Machine Learning Models for Decision-Making, Asymptotic Injury Modelling.

(iv) Performance & Limitations: Metrics: Mean Absolute Error (MAE), RMSE, Judicial Cost Estimation Accuracy

Achieved: High level of accuracy when predicting actual judicial costs in 88 real-life RTA case studies.

Restrictions: Only numerical / judicial dispute costs were considered, not unstructured text policy clause extraction.

Scope for the future: Combining text-mining of exclusion statements of insurance products with financial dispute estimation.

Paper 4: Blockchain technology in transforming Health Insurance Claims Adjudication

Title: Transforming Health Insurance Claims Adjudication with Blockchain Technology; Author: Jaypee Greens; Publisher: Jaypee Press.

Other publications include Procedia Computer Science (Elsevier), 2023.

Publisher: MobiSPC Conference Research Group, LLC.

(iii) Solution: Mobile phones, computers, and other devices can be used to process health insurance claims in a decentralized manner.

The methodology adopted in this project includes: Smart Contracts, Decentralized Ledger

Technology (DLT), Automated Rules Verification.

(iv) Performance & Limitations: Metrics: Claim Settlement Latency, System Security, Throughput (TPS)

Achieved: Claims settlement time decreased from weeks to near real-time automated claims settlement.

Limitations: Does not have natural language processing capabilities that can process raw policy text, only blockchain architecture.

Smart Contracts and NLP policy clause extraction for automated claims: Future Scope.

Paper 5: Explainable AI and Law: An Evidential Survey says that the paper holds the title.Title of the paper is Explainable AI and Law: An Evidential Survey.

The title and author of the paper is: Explainable AI and Law: An Evidential Survey.The paper's title and author are: Explainable AI and Law: An Evidential Survey.

Digital Society (Springer), 2024

Editors: Karen McGregor Richmond, Satya M. Muddamsetty, Thomas Gammeltoft-Hansen, Henrik Palmer Olsen and Thomas B. Moeslund

(ii) Problem & Objectives: Black-box opacities in AI legal decision-making, impacting accountability & transparency.

(iii) Methodology Adopted: Explainable AI (XAI) Taxonomy Mapping

Explainability Fidelity, Stakeholder Trust Score, Interpretability are used to explain performance & limitations: metrics.

Achieved: Developed a structured taxonomy that correlates the legal inference types with clear models that are algorithmic.

Limitations: Conceptual survey framework but no automatic empirical text processing code.

Legal NLP Classifiers with XAI heads: e.g., SHAP/Integrated Gradients.

Paper 6: Legal Information Retrieval and Entailment Using Transformer-based Approaches

Students who have successfully completed Papers 5 and 6 may obtain exemptions from Papers 1 and 2.Students who have passed Papers 5 and 6 can be exempted from Papers 1 and 2.

Legal Information Retrieval and Entailment Using Transformer-based Approaches (i)

The Review of Socionetwork Strategies, 2024 - (Springer)

Published by: The Social Sciences Curriculum Support Center (SSCSC), No. 130 Pinnacle Ave # 210, San Diego, California 92121-1308, United States of America

(ii) Problem & Objectives: Information overload and complex legal entailment verification in case law and statute retrieval (COLIEE).

(iv) The most critical factor remaining to be resolved is the extent of the change.(iv) The most critical question yet to be answered is how much change.

(iv) Performance & Limitations: Metrics: Precision, Recall, F1-Score, COLIEE Task Accuracy

Achieved: High accuracy in entailment tasks in legal paraphrase and condition matching on Legal-BERT-ECHR.

Restrictions: Assesses on statutes; excludes consumer insurance policy contracts.

Future Scope: Premise-Claim verification using legal entailment transformers in Insurance Contracts.

Paper 7: InsuranceRAG: A multi-module retrieval-augmented AI system for intelligent insurance assistance

The title, author, and publisher are as follows:Title: InsuranceRAG; Author: The authors are the members of the Insurance Market Institute of Carolina; Publisher: Insurance Market Institute of Carolina.

Software Impacts (Elsevier), 2026

Sejal Hanmante, Shivani Patil, Priyanka V. Deshmukh and Aniket K. Shahade contributed to this article.

(ii) Problem & Objectives: No transparent, conversational AI to resolve queries and retrieve documents at a clause level in health insurance.

(iv) The methodology that we used is Retrieval-Augmented Generation (RAG), FAISS Vector Store, Domain-adapted Embeddings, and LLM Evaluator.

The performance and scope of the system: Retrieval Recall@K, Answer Faithfulness, Latency, User Evaluation.

Achieved: High retrieval accuracy and understandable clause level responses through a web dashboard.

Restrictions: It's a conversational QA with no ability to compute a structured composite policy claim risk score.

Integrating automated policy risk scoring algorithms with RAG clause extraction: Future scope.

Paper 8: Optimizing Claim Assessment Process for Property Insurance using Machine Learning

Optimizing Claim Assessment Process for Property Insurance using Machine Learning (i)

Procedia Computer Science (Elsevier), 2024.

Publisher: WIT Press

(ii) Problem & Objectives: imbalance issue in data and low feature importance of property insurance claim frequency and risk prediction.

(iv) Evaluation Metrics: Accuracy, specificity, and sensitivity.(v) Results: The results showed that the precision of the Random Forest model with SMOTE Data Imbalance Oversampling is 86%, outperforming the Logistic Regression model with a precision of 80%.

This is a no-fail class and there will be no exams.This is a non-graded course and no examinations will be given.

Without SMOTE balancing, attained: Achieved 96% Recall with Random Forest.

Limitations: Can only work with tabular structured risk features, not with text in the PDF contract.

Future Scope: Post processing with tabular structure ML models and exclusion extraction with NLP from text.

Paper 9: Structure-Aware Uncertainty-Calibrated Evaluation for Legal Contract Knowledge Graph Extraction

we present the structure-aware uncertainty-calibrated evaluation to aid in the evaluation of legal contract knowledge graph extraction.In Paper 9, we propose the structure-aware uncertainty-calibrated evaluation to facilitate the evaluation of legal contract knowledge graph extraction.

This is the very first work to structure-aware uncertainty-calibrated evaluation for Legal Contract Knowledge Graph extraction.It is the first work to structure-aware

uncertainty-calibrated evaluation for Legal Contract Knowledge Graph extraction.

Article in the journal Procedia Computer Science (Elsevier), 2026

Publisher: XDL Press, publisher for David Logub, the musician behind effortlessmixing.com.Publisher: XDL Press - David Logub's EASY MIXING PRESS.

(2) Problem & Objectives: Epistemic uncertainty and inter-judge disagreement evaluation in Legal Contract Knowledge Graph Extraction.

Please note that due to different orders, the results for datasets 0, 4, 8, and 9 may not add up to 100%.Note that the results for data set 0, 4, 8, and 9 do not sum to 100% as orders may have changed.

The last part covers the performance and limitations of the current measuring system to assess the following: (iv) the Coefficient of Variation (CV), performance and limitations of the current measuring system assessed through the AUC for Error Detection, Calibration Error.

Attained: 0.982 AUC for the extraction of contract knowledge graph in terms of error detection.

Limitations: Consumes too much computing resources for multi-judge ensembles; needs reference free validation data.

Lightweight, single-pass real-time contract scanning uncertainty calibration.

Paper 10: On Automated Attribute Extraction from Legal Data with a LLM for a Case Study

A case study for automated attribute extraction from legal documents using large language models (i): Title, Author, Publisher

Artificial Intelligence and Law (Springer)

This article is authored by Subinay Adhikary, Procheta Sen, Dwaipayan Roy and Kripabandhu Ghosh.

(iv) problem & obj: Concept, attribute and tag extraction of fine-grained concepts from unstructured legal proceedings.

Weakly Supervised Sequence Labeling (previously known as SRL): This is the only weak supervision methodology employed for the analysis.Few-Shot Prompting with LLaMA-2 & Black-Box LLMs: The only weak supervision methodology used for the analysis is Few-Shot Prompting with LLaMA-2 & Black-Box LLMs.

(iv) Performance & Limitations: Metrics: Span Extraction Precision, Span Extraction Recall, Macro F1, Downstream Prediction Accuracy

First achieved: LLaMA-2 shows better few-shot attribute generation and downstream statute prediction accuracy.

Limitations: Hallucinations in LLM and sensitivity to prompts in settings where prompts are not well formed.

Application: Future Scope for Fused Few-Shot LLMs for Extraction and Fine-Tuned Transformer Classifier Heads.

Paper 11: Learning Multi-channel Heterogeneous Graphs for Detecting Health-Insurance Fraud.

You can help prevent health insurance fraud by identifying and addressing the unique characteristics of multi-channel heterogeneous graph structure.By recognizing and tackling the specific features of MCHSG, you can effectively combat health insurance fraud.

Heliyon (Elsevier), 2024

Group: PELSRAO - Institute of Electronic Engineering and Microelectronics, Aerospace Institute of Xi'an, and Department of Electronic Engineering, Xidian University

The main problem and the objectives are as follows: (ii) It is difficult to control and detect complex evolving patterns of health insurance fraud in heterogeneous entities like patients, doctors, medicines.

Steps taken were: (iii) Methodology Adopted: Multi-channel Heterogeneous Graph Structured Learning (MHGSL), Graph Convolutional Networks (GCN).

(iv) Performance & Limitations: Metrics: Classification Accuracy, Macro F1, Precision, Recall, AUC-ROC

Achieved: High fraud detection accuracy – captured multi-entity relational topology.

Constraints: Only runs for structured relational graph data, not for unstructured policy document text.

Future Scope: Graph Augmented Text Processing: Policy Exclusions and Claim History.

Paper 12: Addressing the notion of trust around ChatGPT in the high-stakes use case of insurance

In the context of the high-stakes use case of insurance, we will examine the concept of trust with ChatGPT through the lens of the following questions:For the high-stakes use case that is insurance, we will explore the question of trust in relation to ChatGPT under the following questions:

Technology in Society (Elsevier), 2024

These contentions are supported by authors Juliane Ressel, Michaele Völler, Finbarr Murphy and Martin Mullins.

(iii) Problem & Objectives: Trust, governance and financial inclusion hazards with consumer-oriented LLMs in high-stakes insurance use cases.

(iv) Creative Process: System Dynamics Simulation (v) Student Product: A Project Based Knowledge Management System for function deployment.(vi) Student Product: A Project Based Knowledge Management System for function deployment.

Evaluate performance and limitations in terms of metrics, qualitative measurement of the trust metric and risk governance alignment.

Achieved: In place domain-specific regulation for consumer facing LLMs, under the EU AI Act.

Inconveniences: No model implementation – only conceptual commentary paper.

Future Scope: Building explainable and verifiable NLP systems compliant to EU standards aimed by the AI Act transparency provisions.

Paper 13: Legal Eligibility Inference from Text: Constraint Extraction using Pretrained Language Models.

The study evaluates which constraints can be reliably extracted from documents by studying its legal eligibility inference.As a study of legal eligibility inference, it evaluates which constraints can be reliably extracted from documents.

Procedia Computer Science (Elsevier), 2025

Language: Kazakh

Description: (ii) Problem & Objectives: Structured Eligibility Restrictions and Condition-Action Pairings from Legal Texts in an automated manner.

(iv) TODO lists created: Locate the associate homographs for the pre-trained BERT model, Create list of named entities from the system call text, Create list of action words from the condition words, Create list of datasets for conditioning the system call.

(iv) Metrics: (NER Precision, Recall, Constraint Extraction Accuracy) Performance & Limitations

Achieved: Good formalisation of entitlement logic reached in a machine-readable constraint.

Limitations: Only available with eligibility constraints; doesn't run composite policy financial risk scores.

Achieving More Transformation: Constraint Extraction for Insurance Policy Claim Exclusion Weights

Paper 14: Software engineering meets legal texts: LLMs for auto detection of contract smells

Attendees will learn how to leverage LLMs for detecting and rectifying potential "smells" in software contracts within legal texts.

The book ML with Applications (Elsevier) 2024

Word limits: 1,100–1,500 for each author.Deadlines: December 15, 2018.

(iii) Problem & Objectives: Identification of 'contract smells' (ambiguities and other hot spots in legal contracts).

List the methodology adopted—GPT-4 Semi-Automated Labeling, BERT, LEGAL-BERT Fine-Tuning, Single vs Multi-Class Classification.Provide a list of methodology used: GPT-4 Semi-Automated Labeling, BERT, LEGAL-BERT Fine-Tuning, Single vs Multi-Class Classification.

In situations of multiple classes, (iv) aims to help the evaluator understand the performance and limitations in terms of Metrics such as Macro F1, Accuracy, Single-Class vs Multi-Class Precision.

Attained: Fine-Tuned LEGAL-BERT for Single-Class Contract Smell Detection Reaches High Accuracy.

Limitations: Knows broader ambiguities in contracts, not necessarily traps for consumers in insurance policies.

Future Scope: Contract Smells for Insurance Policy Draft Exclusion Traps.

Paper 15: Automated construction contract analysis for risk and responsibility assessment using natural language processing and machine learning

the authors explore the analysis of construction contracts for risk and responsibility assessment using automated technology, specifically natural language processing and machine learning.In Paper 15, the authors delve into the domain of analyzing construction contracts in the context of risk and responsibility assessment, leveraging automated tools such as natural language processing and machine learning.

Natural Language Processing and machines learning to Automate Construction Contract analysis for risk and responsibility assessment (i).

Automation in Construction (Elsevier), 2023

The authors are grateful to Mahmut Ozkan and Nubar Ozkan for their valuable comments and feedback.The authors are thankful for the helpful comments and feedback provided by Mahmut Ozkan and Nubar Ozkan.

(2.) Problem & Objectives: Automated Contract Review to provide Risk & Responsibility Data for preparing Risk Management Plans.

(iii) Methodology Adopted: NLP Vectorization, Machine Learning Text Classification, Ensemble Models, Sentence Type Classifier

(iv) Performance & Limitations: Metrics: Accuracy, Macro F1-Score

Attained: A high classification accuracy score for sentences (Obligation, Risk, Right categories 3,485 sentences).

Clarification: Domain specific to construction contracts (FIDIC) and NO evaluation for health/life insurance policies have been conducted.

Future Scope: Risk/Responsibility sentence categorization for cross domain adaptation to the consumer insurance domain.

4. Summary of Literature, Limitations, and Proposed Methodology:-

The literature reviewed and identified gaps are summarized.

The Gap: Existing contract analysis systems deal with enterprise (corporate) contracts (e.g., M&A), but ignore the fine print of consumer insurance policies.

2. Surface Keyword Failure: Exclusions that are negative framed (e.g., "shall be non payable" or "subject to 24-month moratorium") are not picked up by standard search tools.

3. Black-Box Outputs: The output of the existing risk tools is a single score, with no corresponding page/line numbers or plain-English explanation of the conditions not covered.

Proposed Alternative Methodology

InsureCheck-NLP fills the gaps through its 4-step processing system:

Extract text from policy PDFs using pdfplumber and segment text into structural sections (Exclusions, Waiting Periods, Limits & Deductibles), which is the desired output in stage 1 (Section-Aware PDF Extraction).

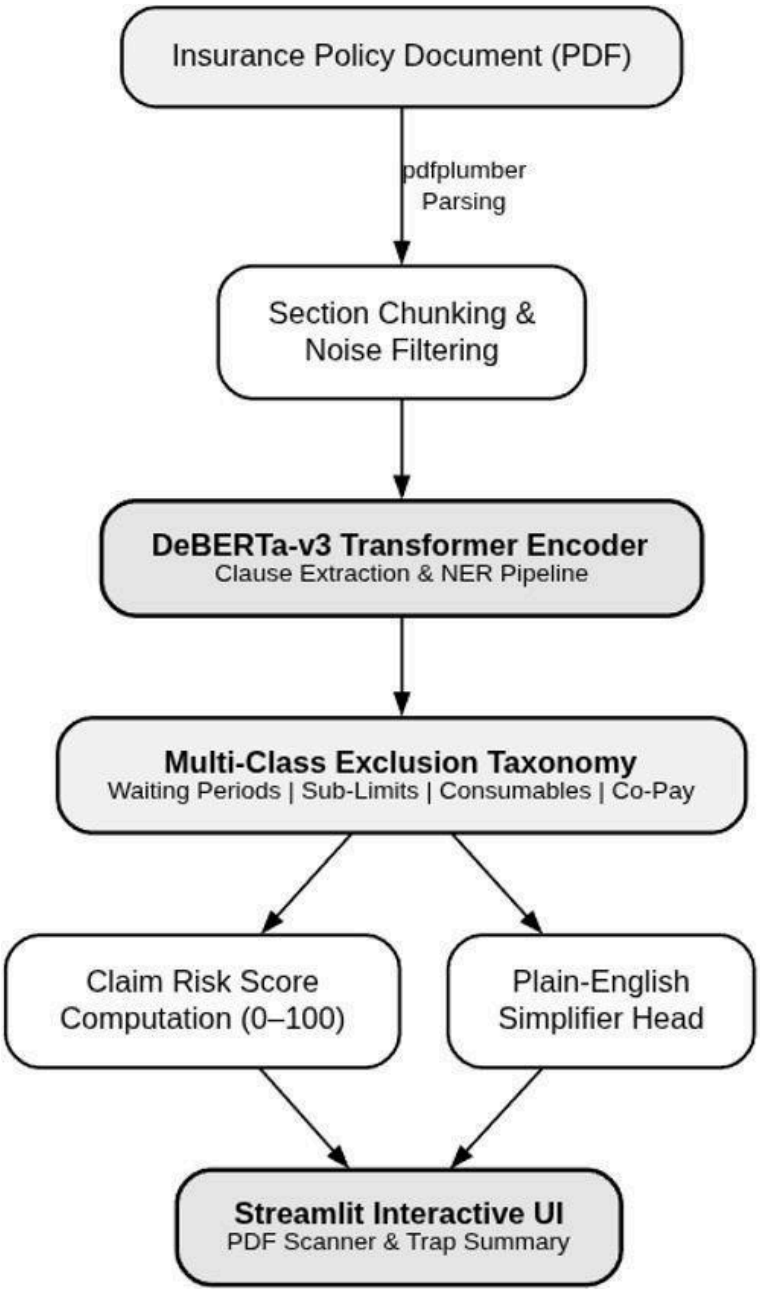
• Stage 2 (Clause Sequence Labeling & Classification): Use a fine-tuned DeBERTa-v3 model to classify exclusions into 4 main categories: Waiting Periods, Sub-Limits & Caps, Non-Payable Items & Consumables, and Co-Payment Mandates.

• Stage 3 (Claim Risk Scoring Engine): Compute a Claim Risk Score

$$R=\min(100,\sum w_i \cdot C_i)$$

Streamlit UI Deployment: Interactive interface created with Streamlit that includes drag and drop PDF upload, interactive Risk Score gauge meter, and simple-English summaries.

ARCHITECTURE DIAGRAM



5. Datasets Description and Sample Data

Datasets Description

1. CUAD (Contract Understanding Atticus Dataset): Legal contract dataset that is public, consisting of 510 commercial contracts with more than 13,000 human-annotated legal clauses across 41 categories.

2. Custom Insurance Policy Corpus: A customized set of 40 real-world consumer insurance policies (Health, Auto, Travel, Life) from the top Indian and foreign insurance providers.

Sample Data & execution example

Raw Policy Text Snippet: "Treatment for cataract has a sub-limit of $300 per eye or 10% of sum insured, whichever is lower, and will not be covered until a minimum of 36 consecutive months of continuous coverage have passed since the start of the policy.

The severity weight for:

Clause 1: Sub-Limit & Cap is 20 (Medium Risk) and the severity weight for Cataract capped at
$300 per eye is 20 (Medium Risk).

Clauses 2: Waiting Period of 36 months continuous coverage for Pre-Existing Diseases are considered to be at 30 (High Risk) and are included in the severity weight. The risk of claim is 50 in 100 (moderate to high risk).

Cataract Surgery Cap: Insurance will cover the maximum amount available of $300 per eye for cataract surgery . The policyholder is responsible for any amount overdue as additional payment due on the extra bill.

3 Year Pre Existing Disease Wait: If you suffer any medical costs for pre-existing disease before you purchase this policy, the 3-year continuity of the policy will exclude the coverage of those costs in the first 3 years.

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